

Assignment

DLL, DML & DCL

Hands-on pgAdmin Practice

Q1. Add Columns (DDL)

Objective: To modify the existing students table by adding two new columns, phone_number and department.

Theory: The ALTER TABLE command is used to modify the structure of an existing table.

The ADD COLUMN clause adds a new column to the table without deleting the existing data.

Here:

- phone_number stores the student's phone number.
- VARCHAR(15) allows variable-length text up to 15 characters.
- department stores the student's academic department.
- VARCHAR(50) allows up to 50 characters.

SQL Solution

```
1 ALTER TABLE academics.students
2 ADD COLUMN phone_number VARCHAR(15),
3 ADD COLUMN department VARCHAR(50);
4
5 SELECT
6     column_name,
7     data_type
8 FROM information_schema.columns
9 WHERE table_schema = 'academics'
10 AND table_name = 'students'
11 ORDER BY ordinal_position;
```

	column_name name	data_type character varying
1	student_id	integer
2	full_name	character varying
3	dob	date
4	email	text
5	cgpa	numeric
6	is_active	boolean
7	skills	ARRAY
8	profile	jsonb
9	enrolled_at	timestamp with time zo...
10	phone_number	character varying
11	department	character varying

Q2. Rename a Column (DDL)

Objective:

To rename the existing full_name column of the students table to student_name.

Theory:

The ALTER TABLE ... RENAME COLUMN command is used to change the name of an existing column.

Renaming a column changes its name while preserving the existing data stored in that column.

SQL Solution:

```
18 ALTER TABLE academics.students
19 RENAME COLUMN full_name to student_name;
20
21
22
23
```

Data Output

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	column_name	data_type
	name	character varying
1	student_id	integer
2	student_name	character varying
3	dob	date
4	email	text
5	cgpa	numeric
6	is_active	boolean
7	skills	ARRAY
8	profile	jsonb
9	enrolled_at	timestamp with time zo...
10	phone_number	character varying
11	department	character varying

Q3. Change a Column's Data Type (DDL)

Objective:

To change the data type of the cgpa column to NUMERIC(4,2) so that values such as 9.50 can be stored accurately.

Theory:

The ALTER COLUMN ... TYPE command is used to change the data type of an existing column.

NUMERIC (4,2) means:

- 4 = maximum total number of digits.
- 2 = number of digits after the decimal point.

For example:

9.50, 8.75, 10.00 can be stored using this data type.

SQL Solution:

```
22 ALTER TABLE academics.students
23 ALTER COLUMN cgpa TYPE NUMERIC(4,2);
24
25
26
```

Data Output

Messages

Notifications

									SQL
	student_id [PK] integer 	student_name character varying (100) 	dob date 	email text 	cgpa numeric (4,2) 				

Q4. Create a New Table (DDL)

Objective: To create a courses table inside the academics schema with appropriate columns and a primary key.

Theory: The CREATE TABLE command is used to create a new table.

A schema is a logical container used to organize database objects such as tables.

The courses table contains:

Column	Data Type	Purpose
course_id	SERIAL	Automatically generated course ID
course_name	VARCHAR(100)	Stores course name
credits	SMALLINT	Stores number of credits
department	VARCHAR(50)	Stores department name

PRIMARY KEY uniquely identifies each course.

SERIAL automatically generates sequential integer values.

SQL Solution

```
6 CREATE TABLE academics.courses(  
7     course_id SERIAL PRIMARY KEY,  
8     course_name VARCHAR(100),  
9     credits SMALLINT,  
10    department VARCHAR(50)  
11 )  
12  
13 --shows table in particular schema  
14 SELECT table_name  
15 FROM information_schema.tables  
16 WHERE table_schema='academics';  
17
```

Data Output Messages Notifications

table_name
name

1 students

2 courses

Q5. Insert Multiple Rows (DML)

Objective:

To insert at least five student records into the students table using a single multi-row INSERT statement.

Theory

The INSERT INTO command is used to add new records to a table.

PostgreSQL allows multiple rows to be inserted using a single INSERT statement. This is more efficient and cleaner than writing separate statements for every row.

Since the table has been modified in previous questions, the values below include all relevant columns.

SQL Solution

```
17 INSERT INTO academics.students
18 (student_id, student_name, dob, email, cgpa, is_active, skills, profile, enrolled_at,
19 phone_number, department)
20 VALUES
21 (1, 'Sunakshi', '2004-06-15', 'sunakshi@gmail.com', 8.90, true,
22 ARRAY['C#', 'SQL', 'React'],
23 '{"city":"Una","year":4,"course":"BE Computer Science"}',
24 '2023-08-15 09:30:00',
25 '9876543210', 'Computer Science'),
26 (2, 'Reetika', '2003-11-22', 'reetika@gmail.com', 8.65, true,
27 ARRAY['Java', 'Spring Boot', 'SQL'],
28 '{"city":"Chandigarh","year":4,"course":"BE Computer Science"}',
29 '2023-08-16 10:15:00',
30 '9812345678', 'Computer Science'),
31 (3, 'Priya Verma', '2004-02-10', 'priya.verma@gmail.com', 9.12, true,
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (100)	dob date	email text	cgpa numeric (4,2)	is_active boolean	skills text[]	profile jsonb
1	1	Sunakshi	2004-06-15	sunakshi@gmail.com	8.90	true	{C#,SQL,React}	{'city': 'Una', 'year': 4, 'course': 'BE Computer Science'}
2	2	Reetika	2003-11-22	reetika@gmail.com	8.65	true	{Java, Spring Boot, SQL}	{'city': 'Chandigarh', 'year': 4, 'course': 'BE Computer Science'}
3	3	Priya Verma	2004-02-10	priya.verma@gmail.com	9.12	true	{Python, Machine Learning, Pandas}	{'city': 'Shimla', 'year': 4, 'course': 'BE Data Science'}
4	4	Vanshika	2003-09-15	vanshika@gmail.com	7.89	true	{C++, DSA, Git}	{'city': 'Delhi', 'year': 4, 'course': 'BE Computer Science'}
5	5	Neha Kapoor	2004-04-10	neha.kapoor@gmail.com	8.27	false	{JavaScript, React, Node.js}	{'city': 'Patiala', 'year': 4, 'course': 'BE Information Technology'}

Q6. Insert Data into courses (DML)

Objective:

To insert at least four course records into the newly created courses table.

Theory

The INSERT INTO statement adds records to a table.

Because course_id is defined as SERIAL, we do not need to manually provide its value. PostgreSQL automatically generates the IDs.

SQL Solution:

```
47
48 INSERT INTO academics.courses
49 (course_id, course_name, credits, department)
50 VALUES
51 (01, 'Data Structures', 4, 'Computer Science'),
52 (02, 'Database Management Systems', 4, 'Computer Science'),
53 (03, 'Computer Networks', 3, 'Information Technology'),
54 (04, 'Operating Systems', 4, 'Computer Science');
55
56
```

Data Output Messages Notifications

	course_id [PK] integer	course_name character varying (100)	credits smallint	department character varying (50)
1	1	Data Structures	4	Computer Science
2	2	Database Management Syste...	4	Computer Science
3	3	Computer Networks	3	Information Technology
4	4	Operating Systems	4	Computer Science

Q7. Update Records (DML)

Objective:

To update the department of all students whose CGPA is greater than 8.0 to Computer Science.

Theory:

The UPDATE command modifies existing records in a table.

The WHERE clause determines which records should be modified.

SQL Solution:

```
57 UPDATE academics.students
58 SET department = 'Computer Science'
59 WHERE cgpa > 8.0;
60
61
62
63
64
65
66
```

		is_active boolean	skills text[]	profile jsonb	enrolled_at timestamp with time zone	phone_number character varying (15)	department character varying (50)
1	7.89	true	{C++,DSA,Git}	{ "city": "Delhi", "year": 4, "course": "BE Computer Science" }	2023-08-18 09:45:00+00	9988776655	Computer Science
2	8.90	true	{C#,SQL,React}	{ "city": "Una", "year": 4, "course": "BE Computer Science" }	2023-08-15 09:30:00+00	9876543210	Computer Science
3	8.65	true	{Java,Spring Boot,SQL}	{ "city": "Chandigarh", "year": 4, "course": "BE Computer Scienc..." }	2023-08-16 10:15:00+00	9812345678	Computer Science
4	9.12	true	{Python,Machine Learning,Pand...	{ "city": "Shimla", "year": 4, "course": "BE Data Science" }	2023-08-17 11:00:00+00	9123456789	Computer Science
5	8.27	false	{JavaScript,React,Node.js}	{ "city": "Patiala", "year": 4, "course": "BE Information Technolo..." }	2023-08-19 12:20:00+00	9090909090	Computer Science

Q8. Delete Records (DML)

Objective:

To delete all students whose `is_active` value is `FALSE`.

Theory:

The `DELETE` command removes existing records from a table.

The `WHERE` condition ensures that only inactive students are deleted.

It is important to use a `WHERE` clause when only specific records should be deleted. Without it, all records would be deleted.

SQL Solution:

```
64 DELETE FROM academics.students
65 WHERE is_active=FALSE;
66
67
68
69
70
71
```

Data Output Messages Notifications

Showing rows: 1 to 4

	student_id [PK] integer	student_name character varying (100)	dob date	email text	cgpa numeric (4,2)	is_active boolean	skills text[]
1	4	Vanshika	2003-09-...	vanshika@gmail.com	7.89	true	{C++,DSA,Git}
2	1	Sunakshi	2004-06-...	sunakshi@gmail.com	8.90	true	{C#,SQL,React}
3	2	Reetiks	2003-11-...	reetika@gmail.com	8.65	true	{Java,"Spring Boot",SQL}
4	3	Priya Verma	2004-02-...	priya.verma@gmail.c...	9.12	true	{Python,"Machine Learn

Q9. SELECT Queries (DML/DQL)

(a) Sort Students by CGPA

Objective: To retrieve all students and display them in descending order of CGPA.

Theory: The SELECT statement is used to retrieve data from a table.

ORDER BY is used to sort query results.

DESC means descending order, so the student with the highest CGPA will appear first.

SQL Solution:

```
69 SELECT * from academics.students
70 ORDER BY cgpa DESC;
71
72
73
74
```

Data Output Messages Notifications						
SQL						
	student_id [PK] integer	student_name character varying (100)	dob date	email text	cgpa numeric (4,2)	is_active boolean
1	3	Priya Verma	2004-02-...	priya.verma@gmail.c...	9.12	true
2	1	Sunakshi	2004-06-...	sunakshi@gmail.com	8.90	true
3	2	Reetiks	2003-11-...	reetika@gmail.com	8.65	true
4	4	Vanshika	2003-09-...	vanshika@gmail.com	7.89	true

(b) Students from Computer Science

Objective: To display only the student_name and email of students belonging to the Computer Science department.

Theory: A SELECT query can retrieve only specific columns instead of all columns.

The WHERE clause filters records according to a condition.

SQL Solution:

```
73 SELECT student_name, email
74 FROM academics.students
75 WHERE department = 'Computer Science';
76
77
```

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SQL	
student_name character varying (100)	email text
1 Vanshika	vanshika@gmail.com
2 Sunakshi	sunakshi@gmail.com
3 Reetiks	reetika@gmail.com
4 Priya Verma	priya.verma@gmail.c...

Q10. Create a Role & Grant Privileges (DCL)

Objective:

To create a PostgreSQL role named `readonly_user` and give it permission to access the `academics` schema and read data from the `students` table.

Theory:

DCL (Data Control Language) is used to control access to database objects.

`CREATE ROLE` creates a new PostgreSQL role/user.

The `LOGIN` option allows the role to log in to PostgreSQL.

`GRANT` gives privileges to a role.

Two privileges are used here:

- **USAGE** on schema:
allows the user to access objects within the schema.
- **SELECT** on table:
allows the user to read records from the table.

SQL Solution:

```
79 CREATE ROLE readonly_user
80 WITH LOGIN PASSWORD 'ReadOnly@123';
81 GRANT USAGE ON SCHEMA academics
82 TO readonly_user;
83 GRANT SELECT ON TABLE academics.students
84 TO readonly_user;
```

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GRANT

Query returned successfully in 85 msec.

Q11. Revoke Privileges (DCL)

Objective:

To remove the SELECT privilege previously granted to readonly_user on the students table.

Theory:

The REVOKE command removes previously granted privileges from a role or user.

After the SELECT privilege is revoked, readonly_user will no longer be allowed to directly read the students table, assuming it has no other source of SELECT permission.

SQL Solution:

```
87 REVOKE SELECT
88 ON TABLE academics.students
89 FROM readonly_user;
90
91
92
93
```

Data Output	Messages	Notifications
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REVOKE

Query returned successfully in 92 msec.

Q12. TRUNCATE Practice (DDL)

Objective: To create an empty copy of the students table, insert test records, remove all records using TRUNCATE, reset the identity/sequence, and verify that the table is empty.

Theory: CREATE TABLE ... LIKE can be used to create a new table with the same structure as an existing table.

The INCLUDING ALL option copies table properties such as columns, constraints, indexes, and other relevant properties.

TRUNCATE TABLE removes all rows from a table quickly.

Unlike DELETE, TRUNCATE is designed to remove the entire table contents rather than individual rows.

RESTART IDENTITY resets identity sequences associated with the truncated table.

SQL Solution:

(a) Create copy and insert test rows

```
1 SELECT * FROM students_copy;
2 CREATE TABLE students_copy
3 (LIKE academics.students INCLUDING ALL);
4
5 INSERT INTO students_copy
6 (student_id, student_name, email, cgpa, is_active, phone_number, department)
7 VALUES
8 (101, 'Test Student 1', 'test1@gmail.com', 8.50, TRUE, '9000000001', 'Computer Science'),
9 (102, 'Test Student 2', 'test2@gmail.com', 7.90, TRUE, '9000000002', 'Information Technology'),
10 (103, 'Test Student 3', 'test3@gmail.com', 9.10, TRUE, '9000000003', 'Computer Science');
```

	student_id [PK] integer	student_name character varying (100)	dob date	email text	cgpa numeric (4,2)	is_active boolean	skills text[]	profile jsonb	enrolled_at timestamp with time zone	phone_number character varying (15)	department character varying (50)
1	101	Test Student 1	[null]	test1@gmail.co..	8.50	true	[null]	[null]	2026-09-05 07:13:58.217812+...	9000000001	Computer Scienc
2	102	Test Student 2	[null]	test2@gmail.co..	7.90	true	[null]	[null]	2026-09-05 07:13:58.217812+...	9000000002	Information Tec
3	103	Test Student 3	[null]	test3@gmail.co..	9.10	true	[null]	[null]	2026-09-05 07:13:58.217812+...	9000000003	Computer Scienc

(b) Truncate the Table and verify

```
15 TRUNCATE TABLE students_copy;
16 RESTART IDENTITY;
17
18
19
20
21
22
```

	student_id [PK] integer	student_name character varying (100)	dob date	email text	cgpa numeric (4,2)	is_active boolean	skills text[]	profile jsonb	enrolled_at timestamp with time zone	phone_number character varying (15)	department character varying (50)
--	-------------------------	--------------------------------------	----------	------------	--------------------	-------------------	---------------	---------------	--------------------------------------	-------------------------------------	-----------------------------------